

Mock exam

Algebra · 60 minutes · formula sheet and calculator allowed

Three questions

Answers · official assessment schedule extracts on the last pages

Instructions

- Attempt all three questions. Show all working.
- You may use the formula sheet and a calculator.
- Time: 60 minutes. Start the clock when you turn the page.
- Each question is graded N0–E8; the total is out of 24.

After the exam

- Mark each question with the assessment schedule: count the u, r and t evidence, then give it a grade from N0 to E8.
- Add the three grades. Cut scores move a little each year: Achievement from about 7–8, Merit 13–14, Excellence 19–20.
- For every mark lost, write the L-number from your handouts next to it.

QUESTION ONE

- (a) Simplify each expression, leaving your answer with positive indices.

(i) $\sqrt[5]{\frac{4(2n)^3}{n^8}}$

(ii) $\left(\frac{n^3}{16n^6}\right)^{-0.5}$

- (b) $\frac{20x^2 - 22x + 6}{5x^2 - 3x}$ can be written in the form $A + \frac{B}{x}$, where A and B are integers.

Find the values of A and B .

- (c) The value, V , in dollars (\$), of a laptop can be modelled by $V = 40 + ke^{-0.5t}$, $t \geq 0$, where t is the time in years since the laptop was purchased. The original price of the laptop was \$900.

How long does it take for the laptop's value to be reduced to 50% of the original value?

- (d) $2y = 2x + 29$ is a tangent to the quadratic $x^2 - 2ky + 32k = 0$, where k is a non-zero constant.

Find the value of k and determine where the quadratic crosses the y -axis.

QUESTION TWO

- (a) (i) Find m if $\log_2(3m + 1) = 4$.

- (ii) Solve the following equation:

$$3 \log_x(64) = 6.$$

- (b) Find an expression for p in terms of x if $\frac{5^{7x+6}}{25^{-x}} = 125^p$.

- (c) Find the value of $6 + \log_b\left(\frac{1}{b^3}\right) + \log_b(\sqrt{b})$.

(d) Using an algebraic method, solve $4^x - \frac{10}{4^x} = 3$.

QUESTION THREE

- (a) (i) Solve the following equation:

$$5x^2 + 13x = 6$$

- (ii) Simplify:

$$\frac{3x^2 - 10x + 8}{9x^2 - 16}$$

- (b) Find the range of values of p for which the graph $f(x) = 2x^2 + 8x + p$ does not cross the x -axis.

- (c) $\frac{x^2 + 2x + k}{x^2 + 7x + 10} = \frac{x - 3}{x + 2}$ where k is a constant.

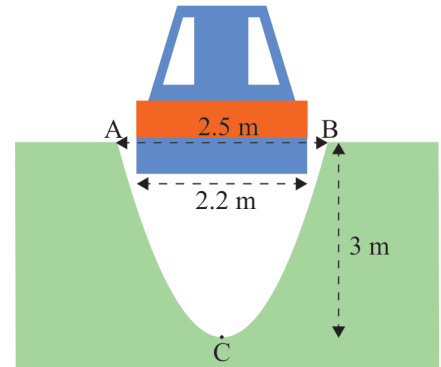
Work out the value of k .

- (d) The diagram below shows the cross-section of a canal and narrowboat floating in the canal.

- The surface of the water (between points A and B) measures 2.5 m across.
- The canal is 3 m deep, at the deepest point C.
- The cross-section of the canal can be modelled as a quadratic curve ACB.
- The cross-section of a narrowboat on the canal can be modelled as a rectangle with a width of 2.2 m.
- The narrowboat must maintain a constant depth below the water of 1 m in order to float.



Source: www.pxfuel.com/en/free-photo-xerdj

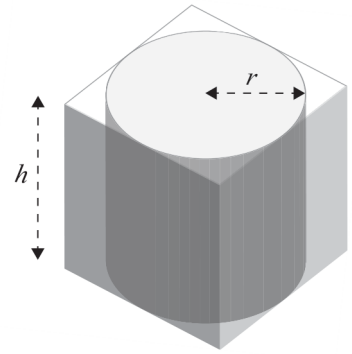


Will the narrowboat be able to float in this canal?

- (e) An open box (i.e. with a base but no lid) has been designed to tightly fit a cylindrical candle. The surface area of the five surfaces of the box is equal to the total surface area of the candle.

Write an expression for the height, h , in terms of the radius, r , and π .

(Surface area of a cylinder = $2\pi r^2 + 2\pi rh$.)



Official assessment schedule extract · Question One

Q	Evidence	Achievement	Merit	Excellence
ONE (a)(i)	$\frac{2}{n}$	Correct simplified expression with positive index, as given.		
(ii)	$\left(\frac{n^3}{16n^6}\right)^{-0.5} = \sqrt{16n^3}$ $= 4n^{\frac{3}{2}}$	- Correct interpretation of negative power. - OR - Interprets power of 0.5 as square root. - OR - Obtains unsimplified equivalent fraction or numerical coefficients such as 2^2 or $\frac{1}{0.25}$.		
(b)	$\frac{2(10x^2 - 11x + 3)}{x(5x - 3)} = \frac{2(2x - 1)(5x - 3)}{x(5x - 3)}$ $= \frac{2(2x - 1)}{x}$ $= 4 - \frac{2}{x}$ $A = 4$ and $B = -2$	Correct factorisation of numerator.	Correct values for A and B (if not explicitly stated, line 3 is required).	
(c)	Initially $t = 0$ $900 = 40 + ke^0$ $k = 860$ $450 = 40 + 860e^{-0.5t}$ $860e^{-0.5t} = 410$ $e^{-0.5t} = 0.477$ Take log of both sides: $\ln(e^{-0.5t}) = \ln 0.477$ $-0.5t = \frac{\ln 0.477}{\ln e}$ $t = 1.48 \text{ years}$	$t = 0$ substituted.	k correctly found using $t = 0$.	Correctly solved.

Official assessment schedule extract · Question One, continued

(d)	$x^2 - k(2x + 29) + 32k = 0$ $x^2 - 2kx + 3k = 0$ using $b^2 - 4ac = 0$ $4k^2 - 12k = 0$ $k = 0$ and $k = 3$ So, $k = 3$, so quadratic can be written as $y = \frac{x^2}{6} + 16$ When $x = 0$, $y = 16$	First step in solving simultaneous equations: substitution for x or y, or equivalent, to give an equation in one variable only OR y -intercept correct	Setting discriminant = 0 for relevant equation (allow minor error) OR Calculus used correctly to obtain both k and y -intercept correctly	T: k correct. TT: both k and y -intercept correct.
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NØ	N1	N2	A3	A4	M5	M6	E7	E8
No response; no relevant evidence.	A valid attempt at one question.	1u	2u	3u	1r	2r	1t	2t

Official assessment schedule extract · Question Two

Q	Evidence	Achievement	Merit	Excellence
TWO (a)(i)	$3m + 1 = 2^4$ $3m = 15, m = 5$	Correct solution.		
(ii)	$\log_x 64 = 2$ $x^2 = 64$ $x = \pm 8$ As base cannot be negative, $x = 8$ OR $x^6 = 64^3$ $x = \sqrt[6]{262144}$ $x = \pm 8$ As base cannot be negative, $x = 8$	Written in an index form. OR $x = 8$ obtained with no consideration of $x = -8$.	Correct answer with justification or evidence of negative value being disregarded.	
(b)	$\frac{5^{7x+6}}{5^{-2x}} = (5^3)^p$ $5^{7x+6-(-2x)} = 5^{3p}$ $9x + 6 = 3p$ $p = 3x + 2$ OR $p = \log_{125} \left(\frac{5^{7x+6}}{25^{-x}} \right)$ $= \log_{125} \left(\frac{5^{7x+6}}{5^{-2x}} \right)$ $= \log_{125} (5^{9x+6})$ $= \log_5 \left(\frac{5^{9x+6}}{3} \right)$ $= 3x + 2$	Conversion to either 5^{3p} or 5^{-2x}. OR $x = \frac{p-2}{3}$ OR Log expression up to line 1, which is only one possible log approach.	Correct answer (simplification not required)	
(c)	$6 + \log_b(b^{-3}) + \log_b\left(b^{\frac{1}{2}}\right) = 6 - 3\log(b) + \frac{1}{2}\log_b(b)$ $= 6 - 3 + \frac{1}{2}$ $= 3\frac{1}{2}$	Combine logs into 1 log term, e.g. $\log_b(b - 2.5)$.	Rewriting both log terms bringing down the power.	Correct value, even if the candidate goes direct to the numerical values

Official assessment schedule extract · Question Two, continued

(d)	$4^x - 10 = 3 \times 4^x$ $4^{2x} - 3 \times 4^x - 10 = 0$ Let $u = 4^x$ $u^2 - 3u - 10 = 0$ $(u + 2)(u - 5) = 0$ $4^x = -2$ or $4^x = 5$ Negative value not valid so: $\log 4^x = \log 5$ $x = \frac{\log 5}{\log 4}$ $x = 1.16$ Accept $\log_4(5)$	• Obtains 4^{2x} OR 16^x OR $(4^x)^2$ OR 3×4^x	• Solved for 'u'.	• Correct value.
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NØ	N1	N2	A3	A4	M5	M6	E7	E8
No response; no relevant evidence.	A valid attempt at one question.	1u	2u	3u	1r	2r	1t	2t

Official assessment schedule extract · Question Three

Q	Evidence	Achievement	Merit	Excellence
THREE (a)(i)	$(5x - 2)(x + 3) = 0$ $x = \frac{2}{5}$ or -3	Both values correct.		
(ii)	$\frac{(3x - 4)(x - 2)}{(3x + 4)(3x - 4)} = \frac{x - 2}{3x + 4}$	Correct expression.		
(b)	Does not touch the x-axis so: $b^2 - 4ac < 0$ $8^2 - 4(2)p < 0$ $64 - 8p < 0$ Accept $p \geq 8$ Could also use completing the square.	- Set up inequality (line 3). - OR $p = 8$ - OR $p < 8$.	Correct answer.	
(c)	$\frac{x^2 + 2x + k}{(x + 5)(x + 2)} = \frac{x - 3}{x + 2}$ $x^2 + 2x + k = (x - 3)(x + 5)$ $x^2 + 2x + k = x^2 + 2x - 15$ Therefore, $k = -15$ Or equivalent approach.	Makes progress towards solution by eliminating denominators, or equivalent.	Value found.	
(d)	Equation $y = a(x + 1.25)(x - 1.25)$ $y = a(x^2 - 1.5625)$ $x = 0, y = -3$ $-3 = -1.5625a$ $a = 1.92$ $y = 1.92(x + 1.25)(x - 1.25)$ $x = 1.1, y = -0.678$ No, boat will not float. OR Using vertex form: $y = a(x - h)^2 + k$ $y = ax^2$ $3 = a(1.25)^2$ $a = 1.92$ $y = 1.92(1.1)^2$ $y = 2.3232$ The edge of the canal is only 0.6768 m below the water surface, so the boat won't float. Accept other correct variations of these approaches.	General equation, fitting the context, formed in any correct format.	Coefficient of x^2 found..	Depth of canal at appropriate width, or width of canal at 1m depth, calculated. AND Statement that boat will not float or similar comment.

Official assessment schedule extract · Question Three, continued

(e)	$2\pi rh + 2\pi r^2 = 8rh + 4r^2$ $\pi rh + \pi r^2 = 4rh + 2r^2$ $\pi h + \pi r = 4h + 2r$ $\pi h - 4h = 2r - \pi r$ $h(\pi - 4) = r(2 - \pi)$ $h = r \frac{(2 - \pi)}{(\pi - 4)}$ $= \frac{r(\pi - 2)}{(4 - \pi)}$ i.e. $h = 1.33r$		Sets up initial equation. OR gives r correctly in terms of h. box is incorrect² OR If surface area of and SA = $Ar + Bhr$, where $B \neq 0$, accept consistent, possibly unsimplified, answer of $h = \frac{r \left(\pi - \frac{A}{2} \right)}{\left(\frac{B}{2} - \pi \right)}$???? indecipherable	T: Correct expression but not simplified. TT: Correct expression.
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NØ	N1	N2	A3	A4	M5	M6	E7	E8
No response; no relevant evidence.	A valid attempt at one question.	1u	2u	3u	1r	2r	1t	2t